Experiment

Creating and Analyzing Disk Images Using dc3dd and Autopsy (Alternative to FTK Imager)

1. Aim

To create a forensic disk image and analyze digital evidence using **dc3dd** and **Autopsy** in **Kali Linux**.

2. Objectives

After completing this experiment, students will be able to:

- Understand the concept of **digital forensics and disk imaging**
- Create a **bit-by-bit forensic copy of a storage device**
- Maintain **data integrity using hashing**
- Analyze disk images to identify **files, deleted files, and system artifacts**

3. Introduction / Theory

Digital Forensics

Digital forensics is the process of **identifying, collecting, preserving, analyzing, and presenting digital evidence** from computers or storage devices.

Digital evidence may include:

- Documents
- Emails
- Browser history
- Logs
- Deleted files

To preserve evidence, investigators create a **forensic disk image**.

Disk Imaging

Disk imaging is the process of creating an **exact replica of a storage device**.

Characteristics of forensic disk images:

- Bit-by-bit copy of storage
- Preserves deleted files
- Maintains data integrity
- Allows investigation without altering original evidence

Hashing in Forensics

Hashing is used to verify the **integrity of digital evidence**.

Common hash algorithms:

Algorithm	Purpose
MD5	evidence integrity
SHA1	verification
SHA256	stronger security

If the hash value of the original disk and the disk image match, the **evidence is considered intact**.

4. Tools Used

Tool	Purpose
Kali Linux	forensic analysis environment
dc3dd	disk image acquisition
Autopsy	forensic investigation and analysis

5. System Requirements

Component	Specification
Processor	Intel i3 or higher
RAM	Minimum 4 GB
Disk	20 GB free space
Software	Kali Linux VM

6. Algorithm

1. Start the Kali Linux environment.
2. Generate activity on the system to create forensic artifacts.
3. Identify the disk partition using system commands.
4. Create a forensic disk image using dc3dd.
5. Generate hash values for integrity verification.
6. Start the Autopsy forensic analysis tool.
7. Import the disk image into Autopsy.
8. Analyze files, logs, and deleted artifacts.
9. Extract relevant digital evidence.

7. Procedure

Step 1: Open Terminal in Kali Linux

Press:

```
Ctrl + Alt + T
```

This opens the terminal.

Step 2: Create Sample Evidence

Generate a file that will later be discovered during forensic analysis.

```
echo "Cybersecurity Lab Evidence" > evidence.txt
```

Verify the file creation.

```
Ls
```

Step 3: Identify Disk Partition

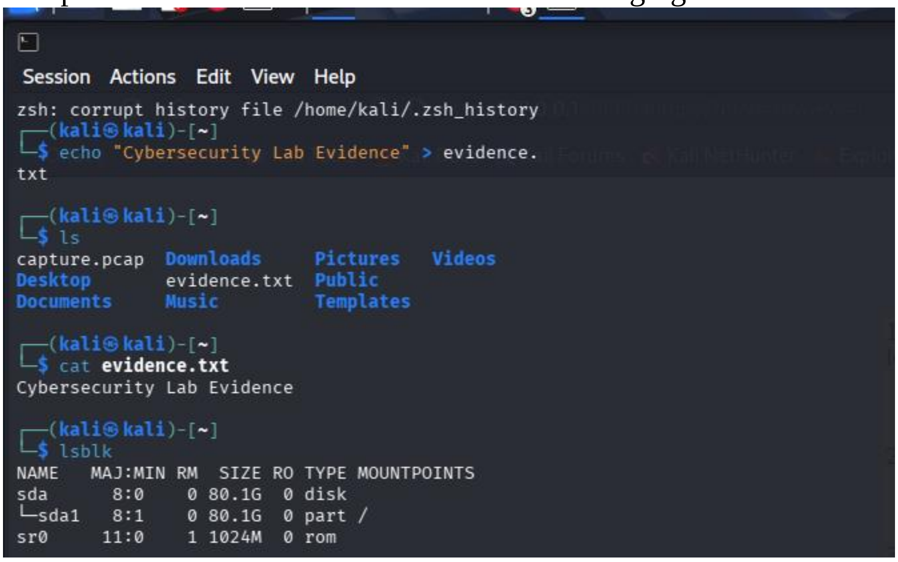
List storage devices.

```
lsblk
```

Example output:

```
sda
 ├─sda1
 ├─sda2
```

The partition **/dev/sda1** will be used for disk imaging



Install dc3dd

```
┌──(kali㉿kali)-[~]
└─$

┌──(kali㉿kali)-[~]
└─$ sudo apt install dc3dd
Installing:
  dc3dd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not U
pgrading: 1757
  Download size: 117 kB
  Space needed: 489 kB / 60.2 GB available

Get:1 http://kali.download/kali kali-rolling/main
 amd64 dc3dd amd64 7.3.1-4 [117 kB]
Fetched 117 kB in 5s (21.9 kB/s)
Selecting previously unselected package dc3dd.
(Reading database ... 454289 files and directorie
s currently installed.)
Preparing to unpack .../dc3dd_7.3.1-4_amd64.deb .
..
Unpacking dc3dd (7.3.1-4) ...
Setting up dc3dd (7.3.1-4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

┌──(kali㉿kali)-[~]
└─$ dc3dd --help
 ______
usage:
 ______

        dc3dd [OPTION 1] [OPTION 2] ... [OPTION N
]

                *or*

        dc3dd [HELP OPTION]

        where each OPTION is selected from the ba
sic or advanced
        options listed below, or HELP OPTION is s
elected from the
        help options listed below.
```

Step 4: Create Disk Image

Step 1: Create a Small Disk File

dd if=/dev/zero of=practice_disk.dd bs=1M count=100

```
┌──(kali㉿kali)-[~]
└─$ dd if=/dev/zero of=practice_disk.dd bs=1M cou
nt=100
100+0 records in
100+0 records out
104857600 bytes (105 MB, 100 MiB) copied, 0.28316
5 s, 370 MB/s
```

Explanation:

Parameter	Description
if	input disk
of	output image file
hash	generate MD5 hash
log	acquisition log file

Creates a **100 MB disk**.

Step 2: Mount the Disk

`mkfs.ext4 practice_disk.dd`

```
┌──(kali㉿kali)-[~]
└─$ mkfs.ext4 practice_disk.dd
mke2fs 1.47.2 (1-Jan-2025)
Discarding device blocks: done

Creating filesystem with 102400 1k blocks and 255
84 inodes
Filesystem UUID: 514f4c47-066d-4a31-8af3-f13b33b7
5b90
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729

Allocating group tables: done

Writing inode tables: done

Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting inf
ormation: done
```

Step 5: Verify Image Creation

Check the created image.

`ls -lh /home/kali/disk_image.dd`

```
┌──(kali㉿kali)-[~]
└─$ ls -lh /home/kali/disk_image.dd
-rw-r--r-- 1 root root 9.8G Mar 15 11:59 /home/ka
li/disk_image.dd
```

Check log file.

`cat /home/kali/acquisition.log`

The log will contain:

- image size
- hash value
- acquisition details

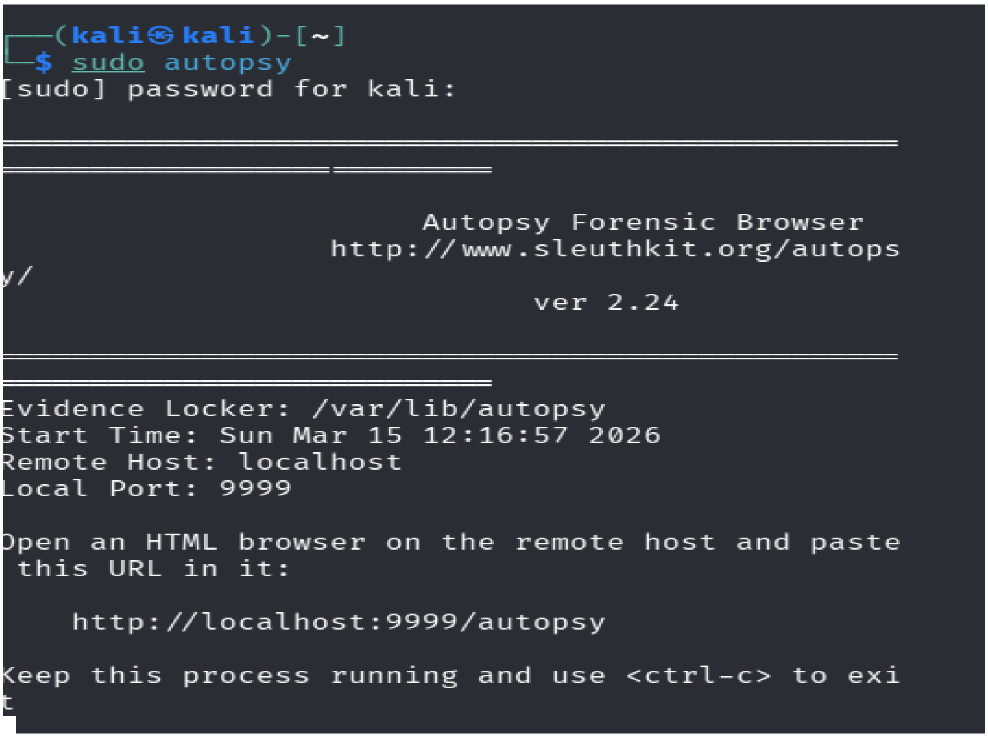
```
┌──(kali㉿kali)-[~]
└─$ cat /home/kali/acquisition.log

dc3dd 7.2.646 started at 2026-03-15 11:52:30 -040
0
compiled options:
command line dc3dd if=/dev/sda1 of=/home/kali/dis
k_image.dd hash=md5 log=/home/kali/acquisition.lo
g
device size: 167966702 sectors (probed),   85,998
,951,424 bytes
sector size: 512 bytes (probed)
 10503356416 bytes ( 9.8 G ) copied ( 12% ), 432.
546 s, 23 M/s

input results for device `/dev/sda1':
   20514368 sectors in
   0 bad sectors replaced by zeros
   ce827a581c4dfce5b976603df85bb5e0 (md5)

output results for file `/home/kali/disk_image.dd
':
   20514368 sectors out

dc3dd aborted at 2026-03-15 11:59:42 -0400
```

Step 6: Start Autopsy

Launch the forensic analysis tool.

`Autopsy` sudo autopsy,, not just autopsy

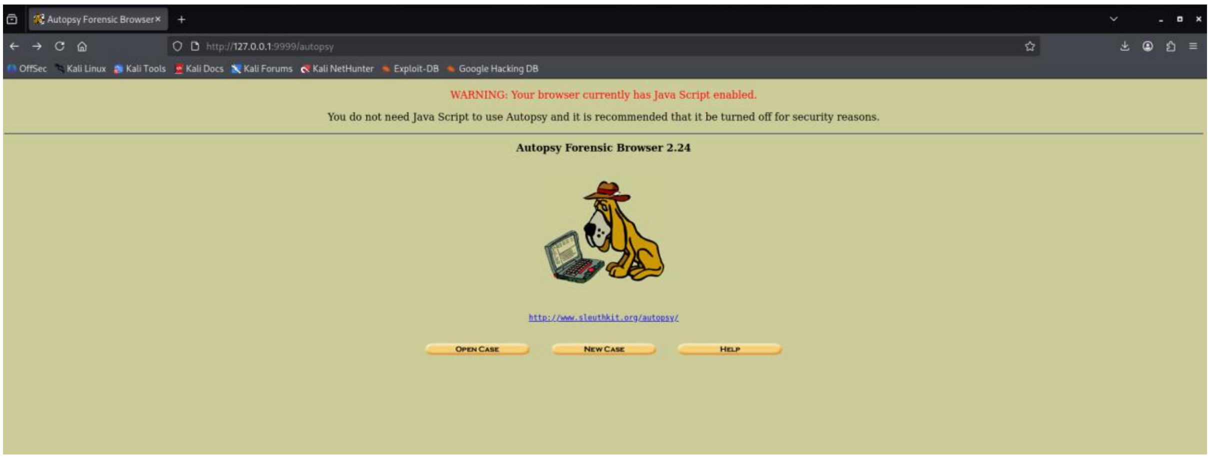
```
┌──(kali㉿kali)-[~]
└─$ autopsy

=================================================
=========================

                       Autopsy Forensic Browser
                  http://www.sleuthkit.org/autops
y/
                             ver 2.24

=================================================
=========================
Evidence Locker: /var/lib/autopsy
Start Time: Sun Mar 15 12:01:18 2026
Remote Host: localhost
Local Port: 9999

Open an HTML browser on the remote host and paste
 this URL in it:

    http://localhost:9999/autopsy

Keep this process running and use <ctrl-c> to exi
```



The terminal will display a URL:

```
http://localhost:9999/autopsy
```

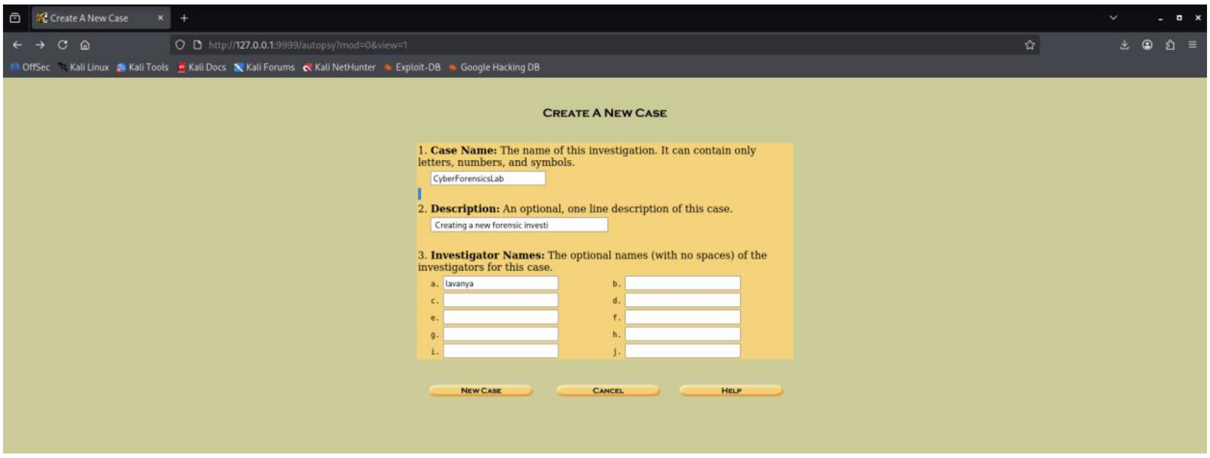
Open this link in the browser.



Step 7: Create a New Case

In the Autopsy interface:

1. Click **Create New Case**
2. Enter case details:



Step 8: Add Disk Image

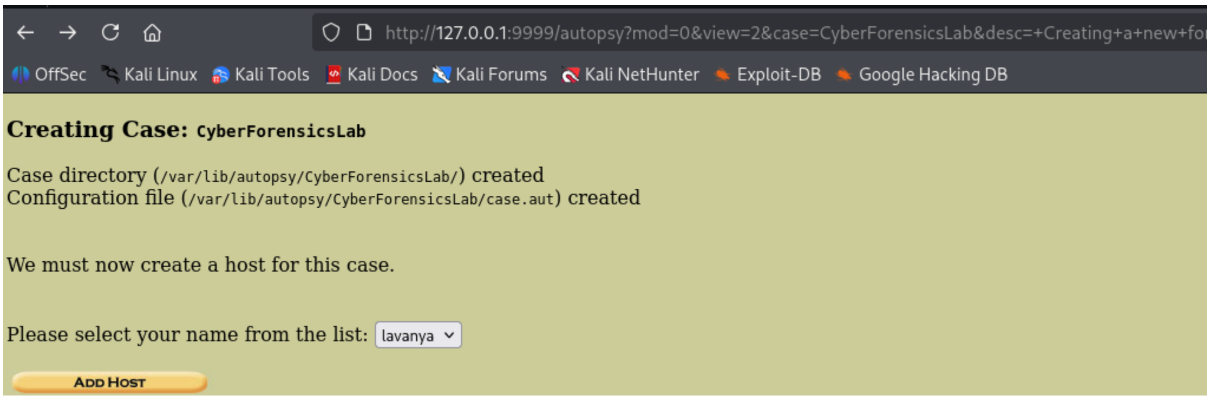
Select:

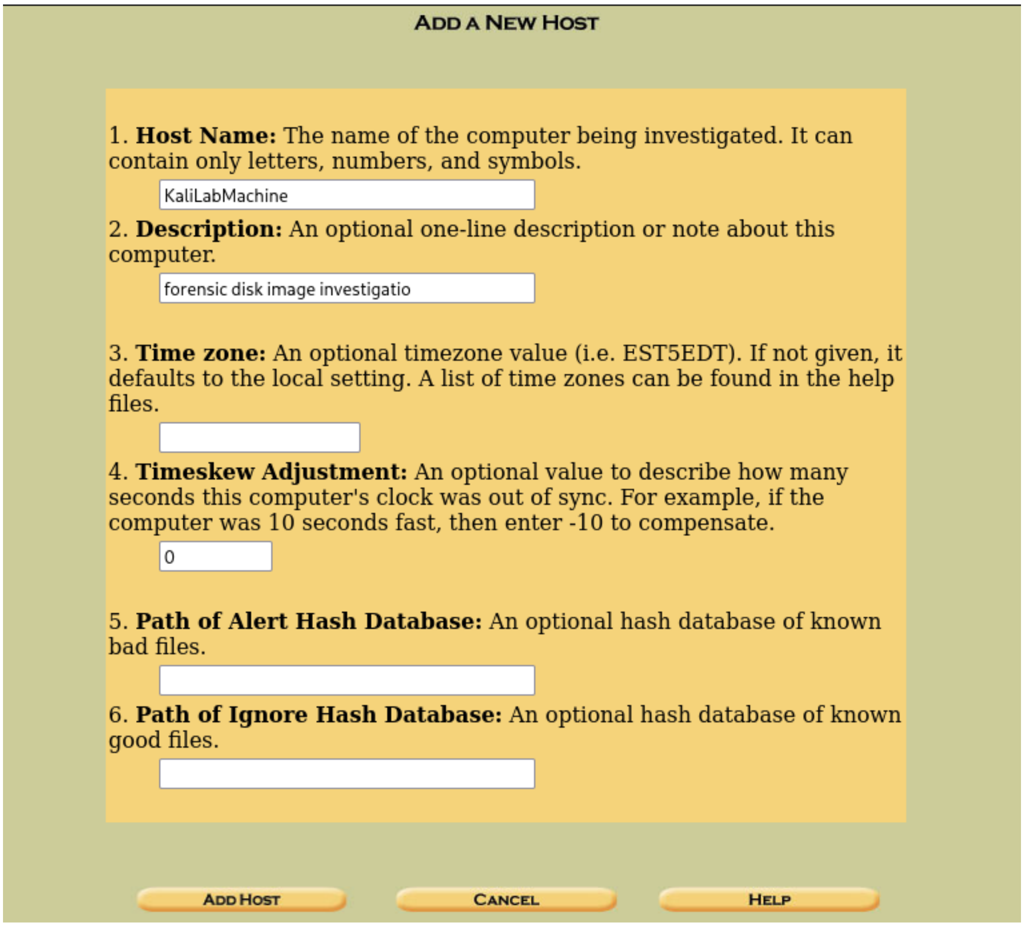
`Add Host → Add Image`

Browse and select:

`/home/kali/disk_image.dd`

Autopsy will begin analyzing the disk image.





Recommended Values

Host Name

`KaliLabMachine`

Description

`Kali Linux virtual machine used for forensic disk image investigation`

Time Zone

`Leave Blank`

Time Skew Adjustment

`Leave Blank`

Alert Hash Database

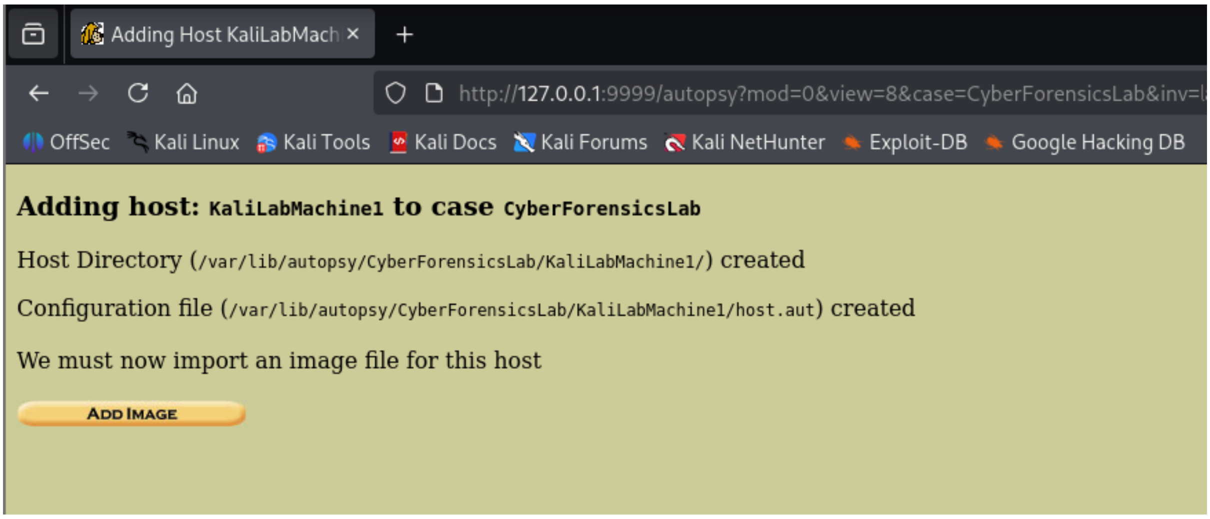
`Leave Blank`

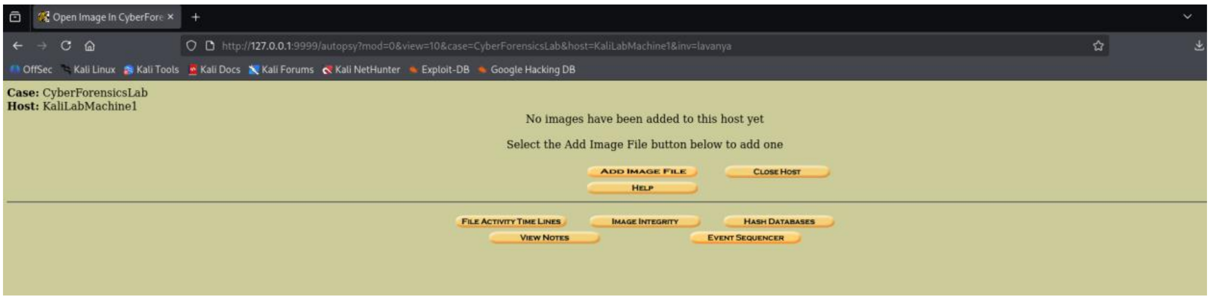
Ignore Hash Database

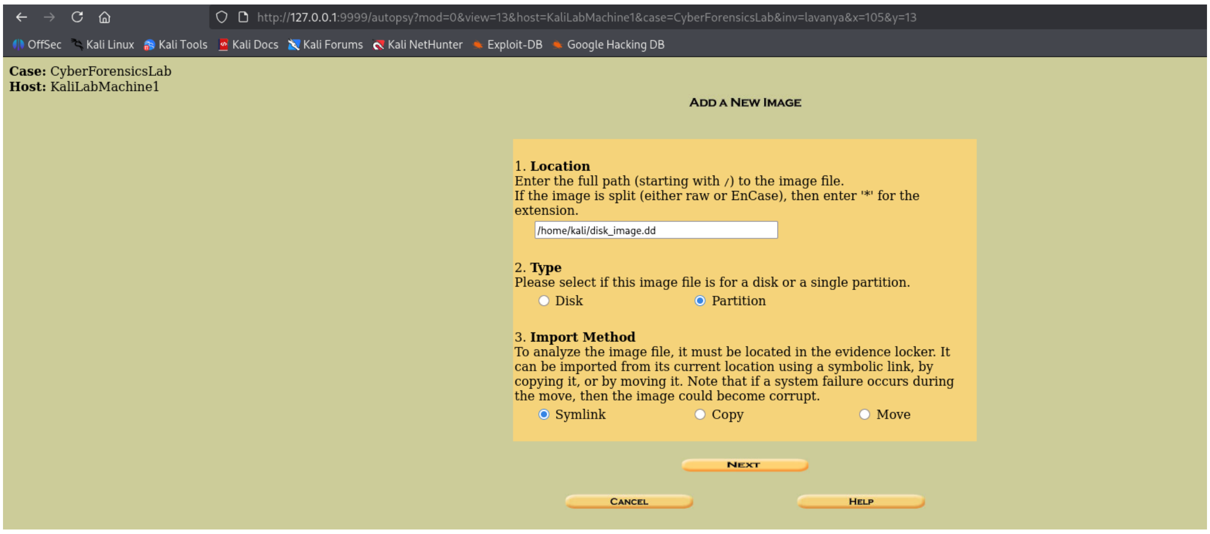
`Leave Blank`

Then click:

Next







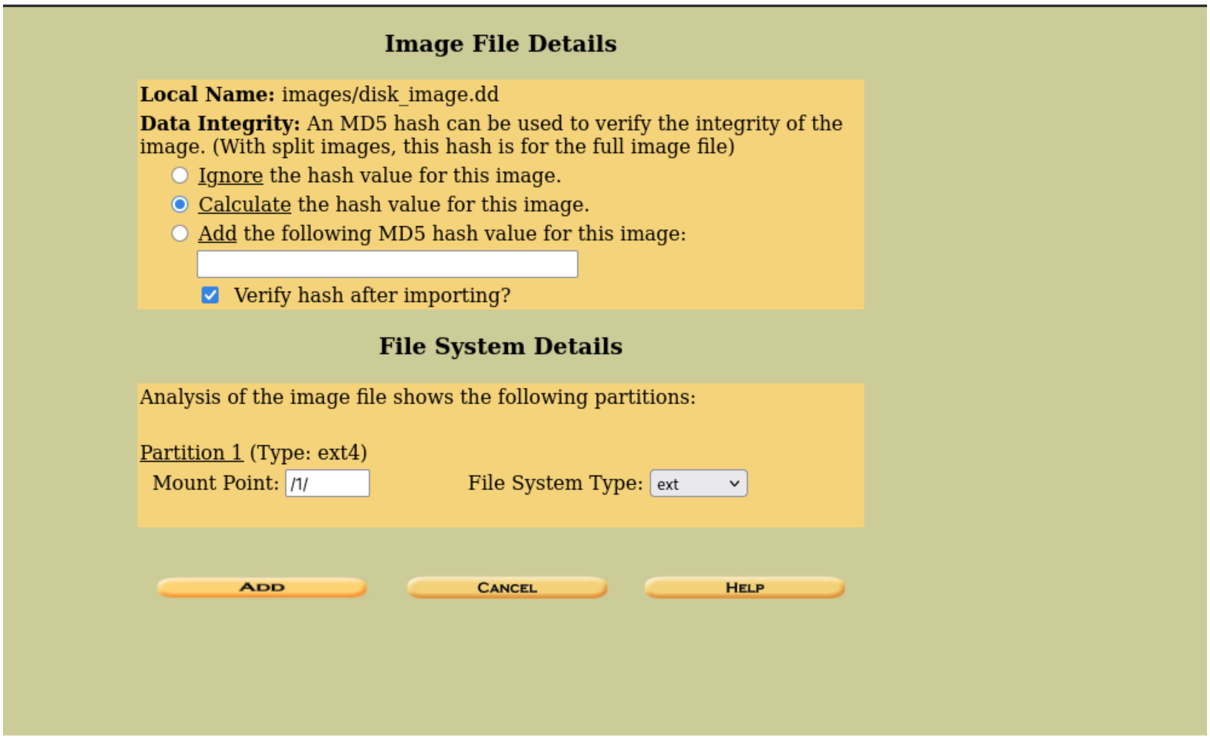
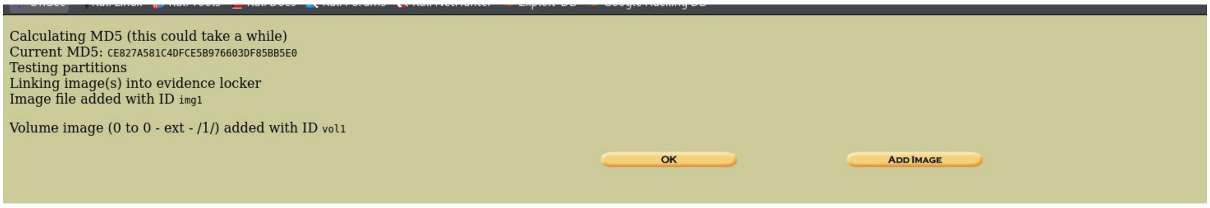
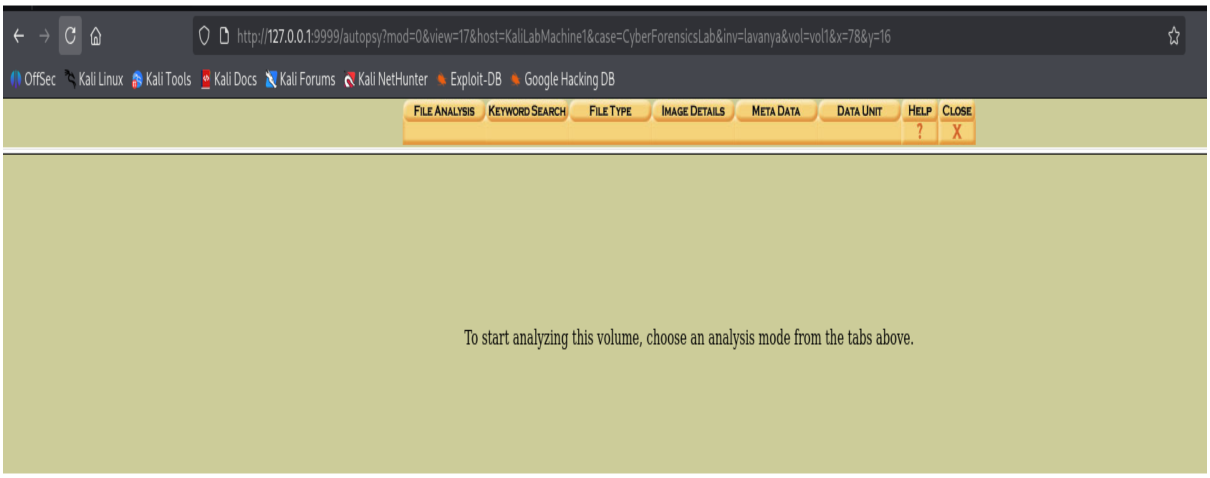
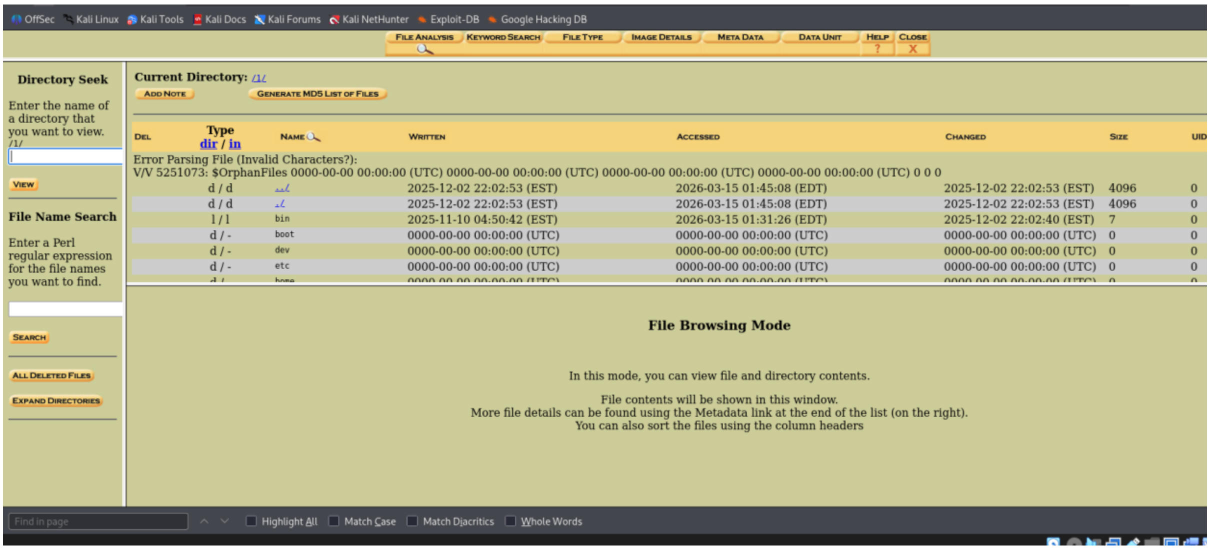


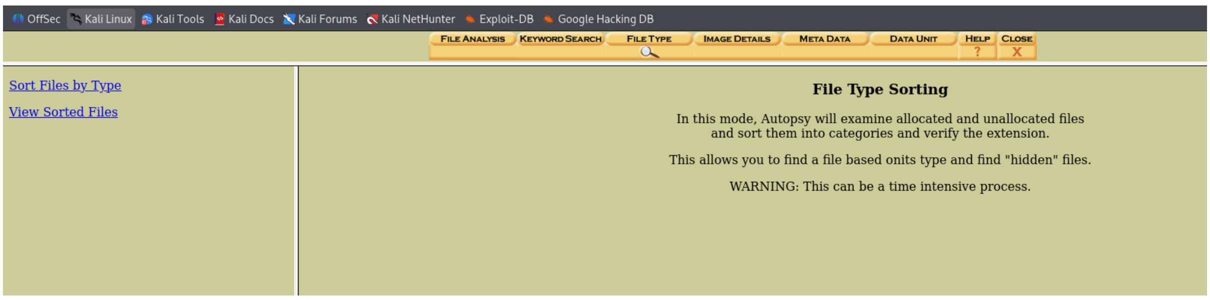
Image File Details
Local Name: images/disk_image.dd
Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)
Ignore the hash value for this image.
Calculate the hash value for this image.
Add the following MD5 hash value for this image:
Verify hash after importing?
File System Details
Analysis of the image file shows the following partitions:
Partition 1 (Type: ext4)
Mount Point: /1/
File System Type: ext
ADD
CANCEL
HELP


Calculating MD5 (this could take a while)
Current MD5: CE827A581C4DFCE5B976603DF85B85E0
Testing partitions
Linking image(s) into evidence locker
Image file added with ID img1
Volume image (0 to 0 - ext - /1/) added with ID vol1
OK
ADD IMAGE

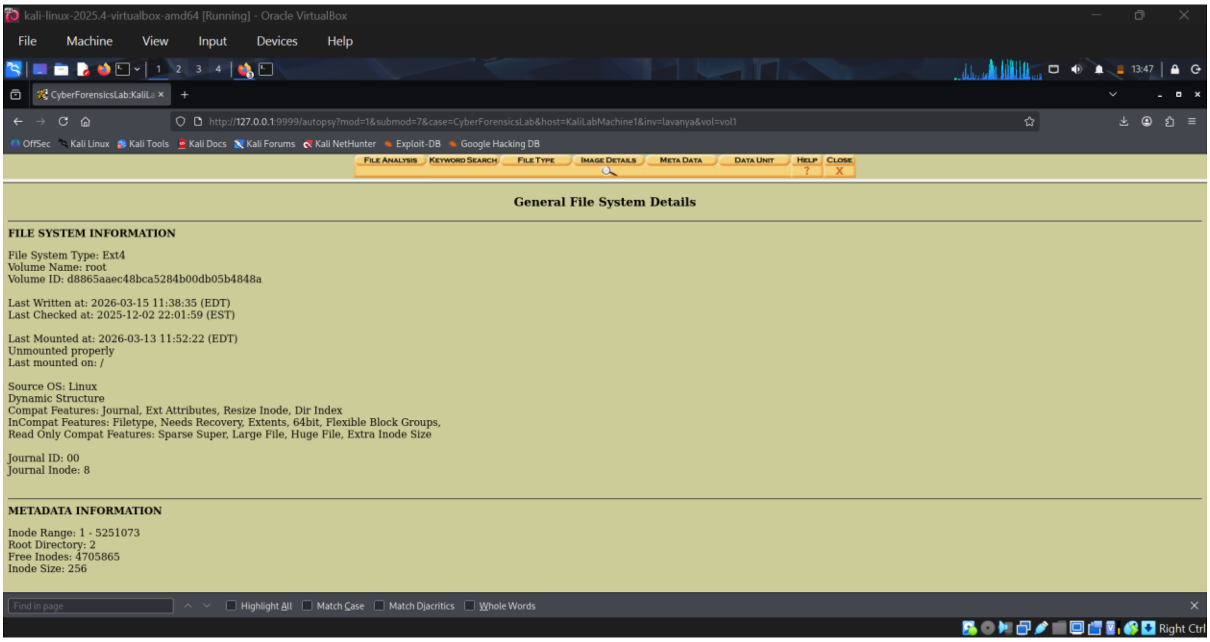

OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
Case: CyberForensicsLab
Host: KaliLabMachine1
Select a volume to analyze or add a new image file.
CASE GALLERY
HOST GALLERY
HOST MANAGER
mount
name
fs type
/1/
disk_image.dd-0-0
ext
details
ANALYZE
ADD IMAGE FILE
CLOSE HOST
HELP
FILE ACTIVITY TIME LINES
IMAGE INTEGRITY
HASH DATABASES
VIEW NOTES
EVENT SEQUENCER


http://127.0.0.1:9999/autopsy?mod=0&view=17&host=KaliLabMachine1&case=CyberForensicsLab&inv=lavanya&vol=vol1&x=78&y=16
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE
?
X
To start analyzing this volume, choose an analysis mode from the tabs above.





Step 9: Analyze Evidence


kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
CyberForensicsLab:KaliLa
http://127.0.0.1:9999/autopsy?mod=1&submod=7&case=CyberForensicsLab&host=KaliLabMachine1&inv=lavanya&vol=vol1
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
FILE ANALYSIS
KEYWORD SEARCH
FILE TYPE
IMAGE DETAILS
META DATA
DATA UNIT
HELP
CLOSE
General File System Details
FILE SYSTEM INFORMATION
File System Type: Ext4
Volume Name: root
Volume ID: d8865aaec48bca5284b00db05b4848a
Last Written at: 2026-03-15 11:38:35 (EDT)
Last Checked at: 2025-12-02 22:01:59 (EST)
Last Mounted at: 2026-03-13 11:52:22 (EDT)
Unmounted properly
Last mounted on: /
Source OS: Linux
Dynamic Structure
Compat Features: Journal, Ext Attributes, Resize Inode, Dir Index
InCompat Features: Filetype, Needs Recovery, Extents, 64bit, Flexible Block Groups,
Read Only Compat Features: Sparse Super, Large File, Huge File, Extra Inode Size
Journal ID: 00
Journal Inode: 8
METADATA INFORMATION
Inode Range: 1 - 5251073
Root Directory: 2
Free Inodes: 4705865
Inode Size: 256
Find in page
Highlight All
Match Case
Match Diacritics
Whole Words
Right Ctrl